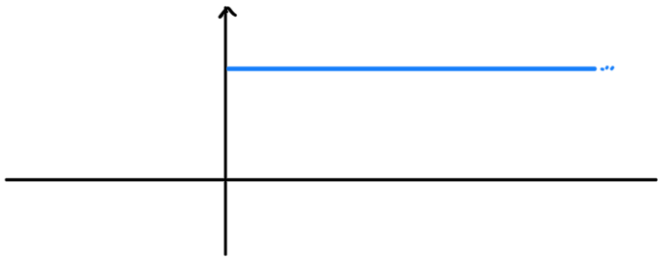
Lezione 2 - 29/02/2024

ESERCIZIO: SI CALCOLINO AREA E VALORE MEDIO DEL SEGNALE $s(t) = 1(t)$ (gradino unitario)



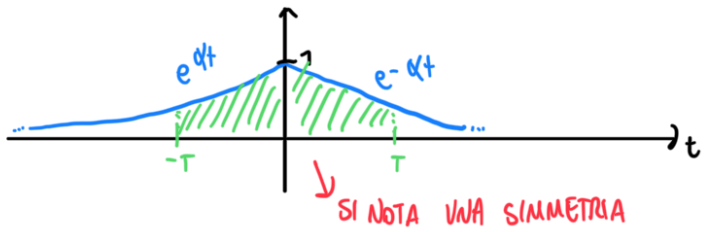
- $A_s = \text{area}(s) = \lim_{T\to+\infty} \int_{-T}^{T} s(t)\,dt = \lim_{T\to+\infty} \int_{-T}^{T} 1(t)\,dt$

siccome per tempi negativi l'integrale è nullo:

$$\lim_{T\to+\infty} \int_0^T 1(t)\,dt = \lim_{T\to+\infty} \int_0^T 1\,dt = \lim_{T\to+\infty} T = +\infty$$ L'AREA È INFINITA

- $m_s = \lim_{T\to+\infty} \frac{1}{2T} \underbrace{\int_{-T}^{T} s(t)\,dt}_{T} = \lim_{T\to+\infty} \frac{1}{2T} T = \frac{1}{2}$

ESERCIZIO 2: SI CALCOLINO AREA E VALORE MEDIO DEL SEGNALE $s(t) = e^{-\alpha|t|}$ $(\alpha > 0)$



$$A_s = \lim_{T\to\infty} \int_{-T}^{T} s(t)\,dt$$

$$= \lim_{T\to+\infty} 2\int_0^T s(t)\,dt = 2\int_0^T e^{-\alpha t}\,dt = \lim_{T\to+\infty} 2\left[\frac{e^{-\alpha t}}{-\alpha}\right]_0^T = \frac{2}{-\alpha}\left(e^{-\alpha T} - 1\right)$$

$$= \lim_{T\to+\infty} \frac{2}{\alpha}\left(1 - e^{-\alpha T}\right) = \frac{2}{\alpha}$$

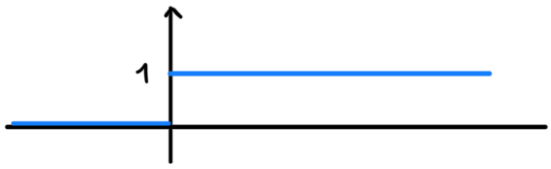
- $m_s = \lim_{T\to+\infty} \frac{1}{2T}\int_{-T}^{T} s(t)\,dt = \lim_{T\to+\infty} \frac{1}{2T}\,\frac{2}{\alpha}\left(1 - e^{-\alpha T}\right) = \underbrace{\frac{1}{T\alpha}}_{\infty}\left(1 - \underbrace{e^{-\alpha T}}_{0}\right) = 0$

ESERCIZIO: ENERGIA E POTENZA DI $s(t) = 1(t)$

$$|s(t)|^2 = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

$\cong 1(t)$

a meno di $t = 0$



$E_s = +\infty$

$P_s = \frac{1}{2}$

- ENERGIA E POTENZA DI $s(t) = e^{-\alpha|t|}$ $\alpha > 0$

$$|s(t)|^2 = \begin{cases} e^{-2\alpha t} & t > 0 \\ e^{2\alpha t} & t < 0 \end{cases} = e^{-2\alpha|t|} = e^{-\beta|t|}$$, DOVE $\beta = 2\alpha$

area e v.m. del modulo quadro

$$A_y = \frac{2}{\beta} = \frac{2}{2\alpha} = \frac{1}{\alpha} \Rightarrow E_s = \frac{1}{\alpha}$$

$$m_y = 0 \Rightarrow P_s = 0$$